

EXECUTIVE BRIEF

The future of enterprise automation: A business leader's guide to agentic AI

TRANSFORMATIONAL CHANGE

15%

of day-to-day work decisions will be made autonomously through AI agents by 2028—up from 0% in 2024[1]

THE FUTURE OF AGENTIC AI

>33%

Gartner predicts more than 33% of enterprise applications will employ AI agents by 2028[2]

[1] "Gartner Identifies the Top 10 Strategic Technology Trends for 2025," Gartner, October 2024
[2] "Intelligent Agents in AI Really Can Work Alone. Here's How." Gartner, October 2024

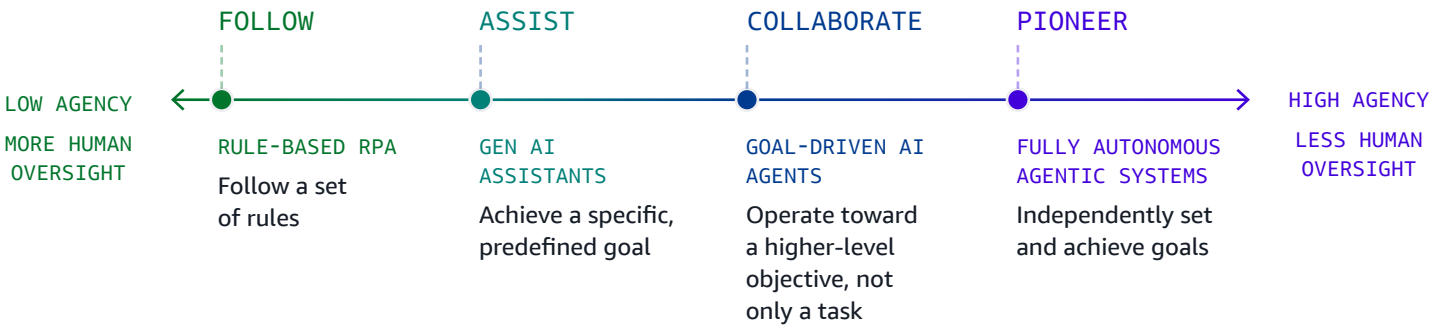
Agentic AI: Reshaping how we work

AI agents are unlocking new enterprise possibilities across industries

The era of AI agents is here. As foundation models (FMs) have grown more sophisticated in their reasoning capabilities and enterprise infrastructure can now support secure, scalable AI deployment, organizations are beginning to tap the full potential of autonomous AI agents. This convergence marks a pivotal moment in AI evolution, enabling intelligent systems that can truly transform how work gets done.

This executive brief will help you understand the business potential of AI agents, evaluate different implementation approaches, and build a strong foundation for autonomous AI success. Plus, get inspired by real-world examples of companies that are fueling innovation with agentic AI on Amazon Web Services (AWS).

Agentic AI maturity scale - from rules to reasoning



What are AI agents?

AI agents are software systems that can reason, plan, and act to accomplish goals with limited human involvement. Unlike gen AI models that respond to isolated prompts, AI agents use memory, reasoning, and planning to execute multistep workflows through thoughtful and adaptable interactions with users, tools, or other systems. They go beyond answering questions to solve complex problems by evaluating results, adjusting their approach, and continuously working toward defined goals—whether that's performing code reviews, processing claims, managing enterprise applications, and more.

How can your business benefit from agentic AI?

4 key areas driving enterprise transformation

In working with businesses across industries to help define their agentic AI strategies, we are seeing autonomous AI agents deliver significant business impact in these areas:

Boosting workplace productivity

AI agents seamlessly integrate with enterprise tools and workflows to automate routine tasks and decision making, dramatically improving operational efficiency.

The NFL leveraged AWS business agents, reducing new hire training time by **67 percent**, and a gen AI–powered assistant so fans can ask questions and get answers to a wide range of questions.[3]

Optimize business workflows

AI agents understand context, adapt to changes, and autonomously manage end-to-end processes, enabling businesses to transform business processes like financial reporting.

Cognizant automated its mortgage compliance workflows with agentic AI to deliver more than **50 percent** in productivity improvements.

Driving research and innovation

AI agents accelerate research and development (R&D) by analyzing data, identifying patterns, testing hypotheses, and generating insights at scale, enabling faster innovation cycles.

Syngenta saw a **5 percent** increase in its crop yields by providing personalized seed recommendations and crop growth modeling to farmers.

Accelerating software development

AI agents can automate code generation, reviews, bug detection, and testing, enabling faster delivery of high-quality code while freeing more time to focus on innovation.

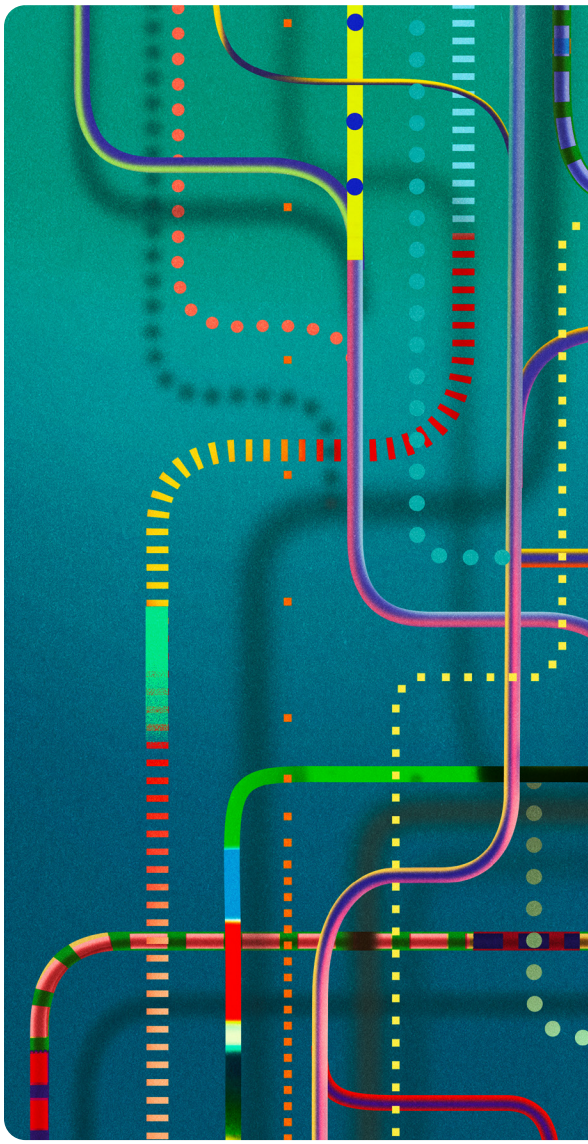
Thomson Reuters used AWS Transform to modernize legacy code bases, cutting costs by **30 percent** and increasing transformation speed by four times compared to manual processes.

[3] "NFL and Amazon Web Services Expand Partnership to Further Shape the Future of Football," National Football League, September 2024

Approaches to implementing intelligent agents

Organizations are discovering that the journey to implementing autonomous agents into business operations isn't one-size-fits-all. And when technology is changing this fast, you need a partner that offers flexibility to quickly evolve and support the path you choose.

We recommend two distinct paths to a successful agentic AI strategy:



1

Specialized agents

For immediate business impact, organizations can deploy pre-built specialized agents designed for specific enterprise functions. Whether enhancing customer service, streamlining IT operations, or optimizing business processes, specialized agents offer the fastest path to value.

2

Custom agents

For businesses with specific requirements or investments in open source frameworks, custom agents provide the flexibility to build tailored solutions while benefiting from enterprise-grade FMs and security controls. This approach combines the innovation potential of open source with the reliability of production-ready infrastructure.

Building your foundation for autonomous AI

Critical areas for enterprise readiness

We believe the key to ensuring your business is ready for agentic AI is through:

Data management

AI agents require seamless access to sources and diverse data types (structured, unstructured, and multimodal) to function effectively. AWS provides the most comprehensive set of services for the entire end-to-end data journey for all workloads, all types of data, and all desired business outcomes, ensuring your data is optimized for agentic AI.

Secure and responsible AI

As AI agents take on greater autonomy and decision making authority, they must operate within established security, ethical, and regulatory frameworks to ensure responsible deployment. AWS is a leader in responsible AI development as the first major cloud provider to receive ISO 42001 certification, demonstrating our commitment to secure and responsible AI systems. AWS also offers automated reasoning checks that can help minimize AI hallucinations and achieve up to 99 percent verification accuracy, helping to ensure agents operate reliably and safely.

Embracing multi-agent environments

As customer needs grow in complexity, single-agent systems often reach their limits. To unlock the full potential of agentic AI, organizations must embrace multi-agent environments where specialized agents work together seamlessly.
Amazon Bedrock AgentCore makes this possible by:

- Enabling you to deploy, and operate AI agents securely at scale using any framework and model.
- Supporting key protocols (Model Context Protocol and Agent-to-Agent) that enable agents to coordinate, share context, and orchestrate tasks across domains, maximizing performance and adaptability while reducing integration friction.

>50%

Over 50% of companies already have AI agents in production, and nearly 80% of businesses have agents in development[4]



Generated using Amazon Nova Canvas "shapes flowing in and out of a funnel."

[4] "State of AI Agents," LangChain, 2024

How businesses are using agentic AI on AWS

Formula 1 reduced issue resolution time by 86% and decreased triage time from days to minutes

In the world of Formula 1 racing, every second counts. When a technical issue occurs during a race, it can take weeks to resolve as engineers investigate the root cause. Using Amazon Bedrock, Formula 1 developed an intelligent root cause analysis (RCA) agent that connects multiple systems to allow engineers to troubleshoot using natural language.

3M leverages the capabilities of Amazon Quick Suite

At 3M, sales teams conduct numerous client meetings and spend excessive time piecing together fragmented information scattered across multiple systems. By leveraging Amazon Quick Suite, an AI-powered workspace, sales teams can quickly get insights through natural conversations with AI agents and automate routine tasks using Quick Flows. This approach is helping drive pipeline progression and opportunities. Additionally, other teams have found value with Quick Research, compressing weeks of intensive analysis into expert-level insights delivered the same day.

Druva, a leading provider of data security, launched enhanced DruAI Agents built on Amazon Bedrock AgentCore, advancing from query-based AI to fully agentic systems.

The agents interpret intent, analyze data, and take autonomous actions—delivering single-command app restoration, instant risk analysis, and secure experimentation environments. Early results show 63 percent of issues are resolved automatically and that escalations resolve 58 percent faster. With zero-trust security, isolated large language models (LLMs), and compliance with FedRAMP and SOC2, Druva is targeting a 70 percent reduction in investigation time and 90 percent automation of routine data protection within a year.

Hundreds of thousands

of customers are moving their gen AI experiments into production on AWS to drive real business value

The AWS vision for AI agents

At AWS, we envision a future where billions of AI agents work alongside humans, accelerating innovation and improving productivity across industries. Our approach balances breakthrough innovation with enterprise-grade reliability—combining cutting-edge AI technology with our proven foundation of security, governance, and scale. This balance of agility and integrity positions AWS as the ideal partner for organizations looking to harness the transformative power of AI agents. Learn more about agentic AI on AWS.

AWS provides everything you need to succeed with agentic AI

PRE-BUILT AGENTS

TOOLS

MODELS

INFRASTRUCTURE

EXPERTISE

OUR VISION

To provide the best place to build and deploy the world's most trusted and useful agents



Fuel your agentic AI breakthroughs with AWS

AWS offers choice and flexibility to accelerate your organization's agentic AI strategy, meeting you where you are on your journey. We provide the most comprehensive set of AI tools and the broadest choice of agents—all with enterprise-proven protection and reliability trusted by millions of customers worldwide.

Whether through custom development, ready-made solutions, or a combination of both, AWS provides the solutions you need to drive business value quickly, making AWS the best place to build and deploy the most trusted and performant agents in the world.

Next steps

Choose your path to agentic AI innovation:

- **Deploy specialized agents**
  Explore how Amazon Quick Suite and Kiro are helping organizations transform productivity and automation

- **Build custom agents**
  Discover how you can build agents with Amazon Bedrock and Strands Agents and deploy and operate them with Amazon Bedrock AgentCore.



Generated using Amazon Nova Canvas "a hummingbird in a garden."

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.